(9) VAD ANTICOAGULATION GUIDELINES

1.0 PURPOSE

1.1 To layout measures and procedures for post-operative anticoagulation of the VAD (Ventricular Assist Device) patient.

2.0 BACKGROUND

To establish guidelines and key notes for post operative anticoagulation management of the paediatric patient on a VAD

3.0 PROCEDURE

3.1. This protocol guides anticoagulation management for patients on durable pulsatile flow ventricular assist devices (Berlin Heart Excor Active) and continuous flow device VADs (e.g., Heartmate III or Centrimag).

3.2. While this is intended as a guide, the VAD team (Cardiology, CTS, CICU and Haematology) may collaboratively decide to make alterations to meet the needs of an individual patient.

3.3. Pre-VAD implantation work-up:

3.3.1. Refer to Haematology team pre-implantation for assessment of baseline thrombotic risk and anticoagulation advice

3.3.2. Document family history of bleeding and thrombotic tendencies. Ask specifically about:

3.3.3. Personal history of unprovoked thrombosis/ extensive thrombosis/ recurrent provoked thrombosis

3.3.4. Family history (in a 1st degree relative) of unprovoked thrombosis, especially in relatives < 40 years of age

3.3.5. Known family history of antithrombin, protein C or protein S deficiency

3.3.6. Prothrombotic screen: Protein C (functional), Protein S (functional), Antithrombin III, Factor V Leiden mutation, Prothrombin G20210A gene mutation, Lupus anticoagulant, Anticardiolipin antibody of immunoglobulin M (IgM) and IgG, Anti-beta 2 glycoprotein 1 antibody of IgM and IgG (at least 2 weeks pre-VAD)

3.3.7. Baseline labs: FBC, Coagulation Profile (aPTT, PT/INR), Fibrinogen (48 hours pre-VAD)

3.3.8. Optional labs: CRP, LDH, ROTEM, where clinically indicated (if pro-inflammatory state is suspected)

3.4. Intra-operative management:

3.4.1. Standard heparin anticoagulation for cardiopulmonary bypass with full protamine reversal in OT

3.4.2. Blood product replacement to normalize coagulation parameters and establish haemostasis in OT.

3.4.3. Meticulous control of bleeding in the operating theatre before returning to the CICU is essential.

3.5 ANTICOAGULATION: **<u>BERLIN HEART</u>**

3.5.1. Background and Overview

Bivalirudin will be the standard anticoagulant used in the early post-operative period in Berlin Heart patients in KKH. It is a direct thrombin inhibitor (DTI) and unlike heparin, does not require antithrombin III (AT3) for activity. Onset occurs almost immediately when given by continuous infusion, with half-life of approximately 20 minutes with normal renal function. Half-life increases with worsening renal impairment. Titration can be done using multiple assays but has most commonly been performed with aPTT. Steady state requires 4 hours.

3.5.2. Following the use of bivalirudin in the acute post-operative phase (POD 0-7), the patient will be transitioned to low molecular weight heparin (LMWH) – subcutaneous enoxaparin for continued anticoagulation while on the Berlin Heart VAD. Dual anti-platelet therapy will also be commenced with feeds.

3.5.3. Post-operative Management: **BIVALIRUDIN**

3.5.3.1. First 6 hours post-op: Establish haemostasis & no anticoagulation

i. Assess for bleeding

ii. Baseline labs upon return to CICU to include: FBC, PT/aPTT/INR, fibrinogen, ROTEM, UECr, LFT, Ca/Mg/PO4

iii. Correct platelets and coagulopathy – aim for platelets > 100K, APTT < 45s, INR < 1.3, fibrinogen > 1.5g/L

3.5.3.2. 6-24 hours post-op: Aim to start BIVALIRUDIN – INITIAL PHASE

Start bivalirudin **<u>only if</u>** the following criteria are met. Ensure there is good haemostasis in the immediate post-op period.

i. Baseline labs have been reviewed

ii. No bleeding: <2mL/kg/hr from chest tubes for at least 4 consecutive hours, and no other sources of active bleeding

iii. Platelets > 100,000

iv. APTT < 45 seconds

v. INR <1.3 and fibrinogen >1.5 g/L

vi. Approval from surgeon and VAD team

3.5.3.3. Consider delaying initiation of bivalirudin (consultant decision) if:

i. Ongoing bleeding (any need for post-operative blood product replacement in last 12h)

ii. Abnormal coagulation profile (elevated INR, elevated aPTT after full protamine reversal, low platelet count) – refer to start criteria above.

3.5.3.4. Relative Contraindications: severe uncontrolled hypertension, severe renal impairment.

3.5.4. Dosage and Monitoring:

3.5.4.1. INITIAL PHASE - BIVALIRUDIN INITIATION

i. For the first 24 – 72 hours postoperatively, patient is at higher risk of bleeding. This is known as the initial phase. During this period, target aPTT is 50-60sec for at least the first 24 hours post-op.

ii. Initial dosing of bivalirudin:

iii. Target aPTT: 50 – 60 seconds.

iv. If renal function is normal (GFR >60ml/min/1.73m2): start IV infusion at 0.3mg/kg/hr

v. In mild-moderate renal dysfunction (GFR 30 - 60ml/min/1.73m2): start IV infusion at 0.2mg/kg/hr

vi. In severe renal dysfunction (GFR <30ml/min/1.73m2): start IV infusion at 0.1mg/kg/hr

3.5.4.2. 2 hours after initiation:

i. Check aPTT

ii. If aPTT is > 60 seconds, reduce dose by half and recheck in 2 hours.

iii. If aPTT is 50 – 60 seconds, keep same dose and recheck in 2 hours.

3.5.4.3. Titrate subsequent doses of bivalirudin using the following table:

TABLE 1: INITIAL PHASE: BIVALIRUDIN TITRATION < 72H POST-OP / HIGH RISK OF BLEEDING
TARGET APTT 50 – 60s

aPTT	Hold	Rate	Recheck
30 – 49	None	Increase by 10%	2 hours after dose change
50 – 60	None	No change	2 hours, then decrease frequency to 4 – 6 hourly if stable
61 – 74	None	Decrease by 10%	2 hours after dose change
75 – 90	*Consider* hold for 15min	Decrease by 20%	2 hours after dose change
> 90	Normal renal function: hold for 15min Mild-moderate renal dysfunction: hold for 45 min Severe renal dysfunction: Discuss with CICU consultant, hold for 2 hours	Decrease by 30% Decrease by 40% Recheck PTT before restarting	2 hours after dose change

3.5.4.4. MAINTENANCE PHASE

i. Subsequent Bivalirudin Titration (after 72 hours post-op): This is known as the **<u>maintenance phase</u>**. Ensure bleeding (i.e chest drain losses) is <0.5ml/kg/hr for at least 6 hours
   a) Maintenance (Standard risk) target aPTT 70 **(goal range 60-80)**
   b) Maintenance (High risk for thrombosis) target aPTT 80 (**goal range 70-90)**

ii. In case of bleeding or thrombosis, the target aPTT may be different. Changes in renal function may affect bivalirudin clearance and may warrant closer monitoring

iii. SIMPLE TITRATION RULE: Adjust the bivalirudin infusion the same % as the difference between the current aPTT and goal aPTT you are trying to achieve. Example: Goal aPTT 75, current aPTT 55 (difference of 20), then increase bivalirudin infusion by 20%.

iv. ** Bolus doses of 0.25-0.5mg/kg have been used to achieve therapeutic aPTT. (Consultant intensivist decision in discussion with VAD team).

TABLE 2A: MAINTENANCE PHASE: BIVALIRUDIN TITRATION > 72H POST-OP
STANDARD RISK – TARGET APTT 60 – 80s

aPTT	Hold/ Bolus	Rate	Recheck
< 50	*Consider* 0.25mg/kg bolus	Increase by 20%	2 hours after dose change
50 - 59	None	Increase by 10%	2 hours after dose change
60 - 80	None	No change	4 hourly (D1 of maintenance dosing) 12 hourly (D2 – 5 of maintenance dosing) Daily (from D5 of maintenance dosing)
81 - 90	None	Decrease by 10%	2 hours after dose change
91 - 100	*Consider* hold for 15 min	Decrease by 20%	2 hours after dose change
> 100	Normal renal function: hold for 15 min Mild-moderate renal dysfunction: hold for 45 min Severe renal dysfunction: Discuss with CICU consultant, hold for 2 hours	Decrease by 30% Decrease by 40% Recheck PTT before restarting	2 hours after dose change

TABLE 2B: MAINTENANCE PHASE: BIVALIRUDIN TITRATION > 72H POST-OP **HIGH THROMBOSIS RISK – TARGET APTT 70 – 90s**			
aPTT	Hold/ Bolus	Rate	Recheck
< 60	*Consider* 0.25mg/kg bolus	Increase by 20%	2 hours after dose change
60 - 69	None	Increase by 10%	2 hours after dose change
70 - 90	None	No change	4 hourly (D1 of maintenance dosing) 12 hourly (D2 – 5 of maintenance dosing) Daily (from D5 of maintenance dosing)
91 - 100	None	Decrease by 10%	2 hours after dose change
100 - 110	*Consider* hold for 15 min	Decrease by 20%	2 hours after dose change
> 110	Normal renal function: hold for 15 min Mild-moderate renal dysfunction: hold for 45 min Severe renal dysfunction: Discuss with CICU consultant, hold for 2 hours	Decrease by 30% Decrease by 40% Recheck PTT before restarting	2 hours after dose change

3.5.4.5. Monitoring while on Bivalirudin:

i. Clinical signs of bleeding
ii. aPTT as shown above
iii. POD 0-7: Daily FBC, UECr, aPTT/PT/INR, fibrinogen
iv. Maintenance lab monitoring depends on patient and device stability. Aim for weekly FBC, UECr, Coagulation profile when stable

3.5.4.6. Administration and Sampling

i. Bivalirudin 250mg powder for reconstitution. Reconstitute with 5mL water for injection to obtain concentration of 50mg/ml. Gently swirl until all material is dissolved
ii. Dilute to 50mL with dextrose 5% or normal saline to give final concentration 5mg/mL
iii. Administer via dedicated central line. Refer to CICU Common Infusion Guide for more information
iv. The infusion rate will be in the first decimal place and dose adjustments will be in the second decimal place.
v. Sampling - aPTTs are to be drawn from heparin naïve arterial or central line strictly.
vi. CVVH: Centres have reported challenges with achieving therapeutic aPTT with continuous haemodialysis.

3.5.4.7. Adverse effects:

i. Bleeding: surgical site, vessel puncture site, gastrointestinal, haematuria, gingival,
ii. Anaemia, thrombocytopaenia
iii. Increased INR: relevant if converting to warfarin
iv. Headache
v. Hypotension
vi. Anaphylaxis
vii. Ecchymosis, skin rash, urticaria
viii. Back pain, groin pain

** In the event that bivalirudin is unsuitable or unavailable for use in a particular patient, discuss with VAD team re alternatives e.g., heparin, argatroban etc.

3.5.5. LOW MOLECULAR WEIGHT HEPARIN: **ENOXAPARIN**

3.5.5.1. Consider switch to subcutaneous enoxaparin (low molecular weight heparin) injection when

i. Completed at least 7 days of bivalirudin infusion and stable
ii. Chest drains and pacing wires are out and no anticipated procedures, or reasons to pause anticoagulation.
iii. No concerns regarding new thrombus formation
iv. Established on maximum dose aspirin and clopidogrel

3.5.5.2. Continue bivalirudin infusion until Anti-Xa (LMWH) is > 0.5

3.5.5.3. Dosage: Enoxaparin subcutaneous twice daily at 0600 and 1800.

3.5.5.4. Send Anti-Xa (LMWH) 4 hours after second dose. Titrate until stable in range.

3.5.5.5. Aim: Anti Xa (LMWH) 0.5 – 1.0. The precise target should be discussed with the VAD team.

i. < 2 months old: 1.7 mg/kg/dose every 12 hours
ii. ≥ 2 months old: 1mg/kg/dose every 12 hours

3.5.5.6. Note: Enoxaparin is renally eliminated and clearance of enoxaparin is decreased by at least 30% in patients with creatinine clearance of <30mL/min/1.73m2

TABLE 4: TITRATION OF ENOXAPARIN		
Anti-Xa level (U/ml)	Dose change	Repeat Anti-Xa
< 0.35	Increase by 25%	4 hours after 2nd dose of new dose regimen
0.35 – 0.49	Increase by 10%	4 hours after 2nd dose of new dose regimen
0.5 – 1.0	Keep same dose	Weekly
1.1 – 1.5	Decrease by 20%	4 hours after 2nd dose of new dose regimen
1.6 – 2.0	● Check <u>trough</u> anti-Xa level immediately before next dose and confirm anti-Xa is <1 U/mL BEFORE dosing at 70% of the last dose. ● Repeat anti-Xa level 4 hours after the 2nd dose of new dose regimen	
> 2.0	● Hold off enoxaparin ● Check <u>trough</u> anti-Xa level (take at the time the next dose would have been due) and repeat anti-Xa level every 12 hourly until anti-Xa level reaches < 0.5 U/mL ● When anti-Xa is <0.5 U/mL, restart enoxaparin at 60% of the last dose. ● Repeat anti-Xa level 4 hours after the 2nd dose of new dose regimen	

3.5.5.7. Converting between Bivalirudin and Enoxaparin (and vice-versa):

i. Continue bivalirudin infusion until LMWH Anti-Xa is > 0.5

ii. Begin bivalirudin no earlier than 8 hours after the last dose of enoxaparin (dose to be discussed with VAD team, guided by aPTT)

a. If starting within 8-12 hours, do not use a bolus dose of bivalirudin

b. If starting after 12 hours, consider a bolus dose of bivalirudin in discussion with VAD team, followed by maintenance dose per protocol.

3.5.6. ANTI-PLATELET AGENTS

3.5.6.1. Aspirin and clopidogrel can be started together as early as 24hrs after commencement of bivalirudin infusion, provided the following conditions are met:

i. No active bleeding concerns – otherwise the option of starting Aspirin alone can be considered

ii. Platelets > 100 x 109/L

iii. Approval by Surgeon & VAD Team

iv. Drains have been removed

v. Feeds are being tolerated

vi. Oral aspirin must be taken after food to reduce risk of gastric irritation.

3.5.6.2. In cases of suspected aspirin / clopidogrel resistance, send blood for platelet aggregation with ADP / arachidonic acid – please consult Haematology for ordering and interpretation of test

TABLE 3: ANTI-PLATELET THERAPY		
High risk of bleeding	PO Aspirin 5mg/kg/daily (max 100mg) alone	
> 50% enteral feeds, good tolerance	PO Aspirin 5mg/kg/daily (max 100mg) PO Clopidogrel 0.2mg/kg (max 75mg)	Note: PO Aspirin dosage range: 5-30mg/kg/day PO Clopidogrel:0.2-1mg/kg/day *Higher doses may be required in select cases where the thrombotic risk is high ** to be discussed with VAD team*
GI concerns	Consider IV Antiplatelet agents if available	

3.6 ANTICOAGULATION: **CONTINUOUS FLOW VAD (e.g., HEARTMATE III or CENTRIMAG)**

3.6.1. Background and Overview

3.6.1.1. Unfractionated heparin will be the standard anticoagulant used in the early post-operative period in patients with a continuous flow VAD (e.g., Heartmate III or Centrimag) in KKH.

3.6.1.2. The activated clotting time (ACT) has been the standard point of care test in our unit to titrate and dose infusions of heparin, in addition to regular aPTT monitoring. The anticoagulant effect of heparin is dependent on an adequate antithrombin level.

3.6.1.3. Onset occurs immediately when given by continuous infusion, with mean half-life of 1.5 hours (range 1-2 hours) with normal renal function.

3.6.1.4. Following the use of heparin in the acute post-operative phase (POD 0-7), the patient will be transitioned to oral warfarin for continued anticoagulation while on a VAD. Anti-platelet therapy (PO aspirin) will also be commenced with feeds and continued for the duration of VAD therapy.

3.6.2. Post-operative Management:

3.6.2.1. 1st 6 hours post-op: Establish haemostasis & no anticoagulation

i. Assess for bleeding

ii. Baseline labs upon return to CICU to include: FBC, PT/aPTT/INR, fibrinogen, ROTEM, UECr, LFT, Ca/Mg/PO4

iii. Correct platelets and coagulopathy – aim for platelets > 100K, APTT < 45s, INR < 1.3, fibrinogen > 1.5g/L

3.6.2.2. 6-24 hours post-op: Aim to start Heparin – INITIAL PHASE
Start heparin only if the following criteria are met. Ensure there is good haemostasis in the immediate post-op period.

iv. Baseline labs have been reviewed
v. No bleeding: <2mL/kg/hr from chest tubes for at least 4 consecutive hours, and no other sources of active bleeding
vi. Platelets > 100,000
vii. APTT < 45 seconds
viii. INR <1.3 and fibrinogen >1.5 g/L
ix. Approval from surgeon and VAD team

3.6.2.3. Consider delaying initiation of heparin (consultant decision) if:

i. Ongoing bleeding (any need for post-operative blood product replacement in last 12h)
ii. Abnormal coagulation profile (elevated INR, elevated aPTT after full protamine reversal, low platelet count) – refer to start criteria above.

3.6.3. INITIAL PHASE

3.6.3.1. For the first 24 – 72 hours postoperatively, patient is at higher risk of bleeding. This is known as the initial phase. During this period, target aPTT is 50-60sec (ACT 160-180) for at least the first 24 hours post-op. Discuss the aPTT goal daily during the VAD round.

3.6.3.2. During the initial phase, check aPTT Q8H.

3.6.4. MAINTENANCE PHASE

3.6.4.1. Subsequent Heparin Titration (after 72 hours post-op):

3.6.4.2. This is known as the maintenance phase. Ensure bleeding is <0.5ml/kg/hr for at least 6 hours

i. Maintenance (Standard risk) target aPTT 70 (goal range 60-80), ACT (180-200)
ii. Maintenance (High risk for thrombosis) target aPTT 80 (goal range 70-90)

3.6.4.3. In case of bleeding or thrombosis, the target aPTT may be different. More frequent POCT with ACT may be done at the bedside.

3.6.4.4. Monitoring of aPTT may be stepped down to Q12H if a steady state has been achieved and desired anticoagulation targets are met.

3.6.5. Dosage:

3.6.5.1. IV heparin infusion is typically commenced at 10units/kg/hr without a bolus dose.

3.6.5.2. Refer to the KKH Guidelines for Anticoagulation Therapy in Children (2023) for titration of heparin dosing. Any change in heparin dose should be followed by a repeat aPTT 4 hours after.

3.6.6. Monitoring while on Heparin:

3.6.1.1. Clinical signs of bleeding

3.6.1.2. aPTT as above

3.6.1.3. POD 0-7: Daily FBC, UECr, aPTT/PT/INR, fibrinogen

3.6.1.4. Maintenance lab monitoring depends on patient and device stability. Aim for weekly FBC, UECr, Coagulation profile when stable

3.6.7 **WARFARIN**

3.6.7.1. Warfarin may be initiated after POD 7 when the following criteria are met:

i. No active bleeding or clotting concerns

ii. Tolerating full feeds with no emesis

iii. aPTT in range and stable for at least 24 hours prior

iv. Approval given by VAD team to transition from heparin to warfarin

3.6.7.2. The target INR is to be discussed with the cardiologist and VAD team (guide range: 2-3) and is decided based on a patient's bleeding and thrombosis risk profile. Refer to KKH Guidelines for Anticoagulation Therapy in Children, April 2023

3.6.7.3. The INR is more likely to remain stable if:

i. Age >6 months of age

ii. No infection

iii. No antibiotics

iv. Consistent daily vitamin K intake from diet, whether eating or fed NG/NJ (Referrals to pharmacist and dietitian for counselling are needed when starting warfarin)

3.6.7.4. Daily INRs are sent until the INR is in the therapeutic range, after which the frequency reduces to every 2-3 days in order to titrate the dose to achieve stability.

3.6.7.5. Bridging bivalirudin with warfarin will need to be discussed with the VAD team if the choice is to switch to PO warfarin in select patients.

3.7 COMMON SITUATIONS WHILE ON ANTICOAGULATION

3.7.1. What do I do if:

3.7.1.1. Patient undergoes a procedure: e.g., chest drain insertion or removal, chest re-exploration etc.

i. Hold bivalirudin for 20 mins prior to procedure.

ii. Restart immediately after procedure at same dose and check aPTT after 2hrs

3.7.1.2. If the patient is scheduled for a long-term central line:
   i. Discuss with IR doctor
   ii. All antiplatelet agents and anticoagulation to continue for non-tunnelled long term lines.
   iii. Avoid tunnelled lines

3.7.1.3. Essential procedures require discussion with all team members to understand bridging plan so that cessation of antithrombotic agents is minimized, while procedural haemostasis is maximized.

3.7.1.4. As a general principle we recommend the following:
   i. For procedures deemed low risk of bleeding:
   Continue all antiplatelet agents, and dose reduce anticoagulation agent for lower therapeutic target around the procedures
   ii. For procedures deemed high risk of bleeding
      a) Hold antiplatelet agents around 3-5 days prior to procedure
      b) Hold bivalirudin ~ 1–4 h prior to skin incision (depending on renal function)
      c) Hold unfractionated heparin 4 h prior to skin incision
      d) For warfarin (Vitamin K Antagonist), start holding 3 days prior to procedure with daily INR measurement. Once INR <2, then start LMWH or continuous infusion of DTI or UFH
      e) For LMWH, withhold dose 24 hours prior to procedure (night before and morning of procedure). Consider bridging with UFH or DTI depending on thrombotic risk of pump and patient.

3.7.1.5. Prompt re-initiation of anticoagulation is required once haemostasis achieved and in discussion with all team members to ensure risk of re-bleeding is minimized, with hourly pump examinations for paracorporeal pumps for 24 h post procedure

3.7.2. Pump thrombus seen

3.7.2.1. Notify the CICU consultant

3.7.2.2. Keep the patient NBM

3.7.2.3. Request a review by the consultant surgeon and VAD team

3.7.2.4. If established on long term anti-coagulation ascertain with the VAD team if these are therapeutic

3.7.3. Patient starts vomiting

3.7.3.1. Suspect poor absorption of enteral antiplatelet or anticoagulation drugs e.g., warfarin

3.7.3.2. Re-check efficacy of anticoagulation (discuss with haematology)

3.7.3.3. Re-examine the Berlin Heart for new thrombus formation

3.7.3.4. Consider conversion to IV antiplatelet therapy alongside IV bivalirudin

3.7.4. Clinical Bleeding while on VAD

3.7.4.1. Children on a VAD may experience minor/major bleeding episodes requiring closer monitoring and adjustment of anticoagulation.

3.7.4.2. Inform VAD team before any changes to anticoagulation/ antiplatelet therapy is made.

3.7.4.3. Refer to ACTION BLEEDING protocol

3.7.5. Patient is going for a transplant

3.7.5.1. Ensure FBC, clotting profile and GXM are sent immediately

4.0 REFERENCE

4.1. Berlin Heart North America. (2020). Advanced Cardiac Therapies Improving Outcomes Network (Action) Direct Thrombin Inhibitor (DTI) Harmonization Protocol for VADs. [online] Available at http://www.berlinheart.com/medical-professionals/anticoagulation/

4.2. Freeman Hospital Paediatric VAD Service Anticoagulation Guidelines 2022

4.3. GOSH Anticoagulation for Berlin heart Excor (Nov 2020)

4.4. Vanderpluym CJ et al. Utilization and Outcomes of Children Treated with Direct Thrombin Inhibitors on Paracorporeal Ventricular Assist Device Support. ASAIO Aug 2020; 66(8):939-945

4.5. Zafar et al. Berlin Heart EXCOR and ACTION post-approval surveillance study report. The Journal of Heart and Lung Transplantation, April 2021,40(4): 251-259

4.6. KKH Guidelines for Anticoagulation Therapy in Children, April 2023

4.7. Ghbeis MB, Vander Pluym CJ and Thiagarajan RR (2021) Hemostatic Challenges in Pediatric Critical Care Medicine—Hemostatic Balance in VAD. Front. Pediatr. 9:625632. doi:0.3389/fped.2021.625632

4.8. ACTION Network: Bleeding Protocols for VAD

5.0 SPECIAL INTRUCTION

To reach out to the VAD team for any further queries